



Local sliced Wasserstein feature sets for illumination invariant face recognition

Yan Zhuang^{a,b,*}, Shiyong Li^c, Mohammad Shifat-E-Rabbi^c, Xuwang Yin^d,
Abu Hasnat Mohammad Rubaiyat^d, Gustavo K. Rohde^{c,d}

^a Windreich Department of Artificial Intelligence and Human Health, Icahn School of Medicine at Mount Sinai, New York, NY 10029, USA

^b Department of Diagnostic, Molecular and Interventional Radiology, Icahn School of Medicine at Mount Sinai, New York, NY 10029, USA

^c Department of Biomedical Engineering, University of Virginia, Charlottesville, VA 22908, USA

^d Department of Electrical and Computer Engineering, University of Virginia, Charlottesville, VA 22908, USA

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ABSTRACT

We present a new method for face recognition from digital images acquired under varying illumination conditions. The method is based on mathematical modeling of local gradient distributions using the Radon Cumulative Distribution Transform (R-CDT) (Kolouri et al., 2015). We demonstrate that lighting variations cause certain types of deformations of local image gradient distributions which, when expressed in R-CDT domain, can be modeled as a subspace. Face recognition is then performed using a nearest subspace method in R-CDT domain of local gradient distributions. Experimental results demonstrate the proposed method outperforms other alternatives in several face recognition tasks with challenging illumination conditions. Python code implementing the proposed method is publicly available, which is integrated as a part of the software package PyTransKit.

1. Introduction

Automated face recognition is a necessary task for many machine-human interaction applications. Illumination variations can cause significant appearance changes for the same person and significantly affect recognition accuracy. Several pioneer studies observed that variations among images of the same person owing to variable lighting are can appear to be larger than those owing to change in identity [1]. One example is illustrated by the well-known Yale B Extended Face database [2], as shown in Fig. 1. The top row of Fig. 1 shows images of the same face acquired with different lighting conditions. The middle row shows image patches near the eye. The bottom row shows the corresponding local 2d discrete gradient distribution within local patches, which change dramatically due to varying illumination conditions. As the figure shows, identifying a person when illumination changes are drastic can be challenging.

Researchers have investigated many approaches to address illumination issues for face recognition. These approaches can be broadly categorized into three sub-types: invariant feature extraction, 3D face modeling, and data augmentation. Some methods can include elements from more than one category. Illumination-invariant feature extraction

methods aim to eliminate lighting effects by holistic decomposition [3], quotient models [4,5], or logarithm difference [6], for example. These approaches apply image processing techniques to “normalize” images in such a way that they have robust appearances under varying illumination. Advantages of these methods include ease of implementation and computational efficiency. Often times, however, the formulation of these local image feature sets are empirical and are often lacking in theoretical understanding with respect to illumination effects. Another type of method uses multiple face images acquired under different illumination conditions to build a generative 3D face model that can approximate and render lighting variations of the face [2,7–9]. However, these methods require many training images with different lighting conditions to construct such a generative model. More recently, deep learning methods have been employed for face recognition [10–13]. These end-to-end solutions leverage a large number of training images and can result in increased performance in standard face recognition tasks [10]. However they typically require a large amount of data to be effective. When only a limited number of training samples are available, one may use data augmentation approach (e.g., brightness and contrast variations [14]) to expand the training samples for deep learning-based

* Corresponding author at: Windreich Department of Artificial Intelligence and Human Health, Icahn School of Medicine at Mount Sinai, New York, NY 10029, USA.

E-mail addresses: yan.zhuang@mssm.edu (Y. Zhuang), sl8jx@virginia.edu (S. Li), mr2kz@virginia.edu (M. Shifat-E-Rabbi), xy4cm@virginia.edu (X. Yin), ar3fx@virginia.edu (A.H.M. Rubaiyat), gustavo@virginia.edu (G.K. Rohde).

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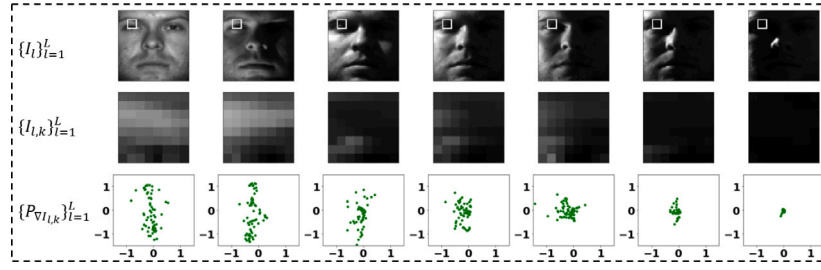


Fig. 1. Top row shows face images $\{I_l\}_{l=1}^L$ under various real-world illumination conditions. Middle row shows k_{th} patches of face images $\{I_{l,k}\}_{k=1}^{k_{th}}$. Bottom row shows that local 2D discrete gradient distributions $\{P_{\nabla I_{l,k}}\}_{k=1}^{k_{th}}$ experience a combination of several types of deformations such as translation, scaling, skewing, rotation, and nonrigid deformations when illumination intensity variation is present.

approaches. However, as demonstrated in the experiments below, the performance improvements are still somewhat limited, especially when illumination effects are severe.

Local image gradient-based feature sets have been used for pattern recognition including face recognition. Numerous methods following this approach have been developed such as SIFT [15], HoG [16,17], and LBP [18]. The idea behind these is to first divide images into small local patches and then compute corresponding representations regarding local gradient information within each local image patch. These low-level local gradient-based descriptive feature sets are favored for face analysis, in part because the spatial differentiation operation naturally eliminates additive constants (i.e. brightness changes). Furthermore, splitting a face image into patches allows classification methods to be more robust with respect to slight misalignment [15,16]. In addition, the assumption that illumination variations in neighboring regions is smooth is widely used for lighting-invariant face analysis [3–6,19]. Thus local low-level features are capable of providing robustness to illumination variations [20].

Our work presented here follows the same line of reasoning. We leverage local patch-wise image gradient measurements to form a new transport-based image descriptive feature set. To be precise, our proposed method divides an image into multiple local image patches, computes the corresponding local image gradients for each image patch, and explicitly represents the local image gradients as 2D discrete distributions. We illustrate that varying illumination conditions lead to certain types of deformations of local 2D discrete distributions within an image patch. Taking this knowledge into account, we construct a novel local low-level image descriptor using a sliced-Wasserstein metric computed with the aid of the Radon Cumulative Distribution Transform (R-CDT) [21,22]. Using certain ‘convexifying’ properties of the R-CDT transform, we are able to build a classifier that is invariant to certain deformations of the gradient distribution caused by changing illumination conditions. We show mathematically that the local (patch-wise) image gradient distribution observed under certain illumination variations forms a convex set in R-CDT space and can thus be linearly separated. Equipped with these mathematical properties, we construct a subspace learning-based classifier to perform illumination-invariant face recognition. Experiment evaluations demonstrate that the proposed method achieves competitive performance among comparable approaches in three different face illumination datasets.

2. Related work

As mentioned above, face classification methods robust to illumination changes can be classified into three main types of approaches: (1) illumination invariant feature extraction, (2) 3D face modeling, and (3) deep learning approaches with data augmentation. Most lie on the spectrum of illumination invariant feature extraction, which refers to approaches to “remove” illumination effects using illumination normalization [6], holistic decomposition [3], quotient models [4,5], and logarithm difference. One straight forward approach is to employ

the log transform for illumination normalization, followed by lighting invariant feature extraction [3,23]. More specifically, Zhu et al. first apply the log transform and then compute HoG features for face recognition [23]. Similarly, Chen et al. use the log transform and then decompose the image into high-frequency and low-frequency components keeping only high-frequency components for face recognition [3]. Lai et al. calculate the difference between neighboring pixel values in log transformation domain, formulating the logarithm-difference edge map. Then based on the size of the neighborhood, different scales of edge maps are computed and aggregated to represent each face [6]. Other approaches such as WebberFace [5] and GradientFace [4] are quotient-based models. To be precise, quotient models seek to represent images in such way that the current pixel value in the new representation is the ratio between the difference of current pixel and its neighboring pixel to the current pixel value in original image. However, although these methods achieve excellent results on some datasets, they are often not effective for illumination variations that include shadows. It is worth noting that most of these approaches focus on using gradient information by directly computing differences between local neighboring pixels [4,23], demonstrating that the effectiveness of gradient information for illumination invariant face representation. In addition, as multiple methods calculate representations in a neighboring region (image patch), they also implicitly rely on the assumption that the illumination variations are locally smooth.

Another research direction is to use a set of images acquired under varying lighting conditions to build a 3D face model that can render all possible illumination variations. Several researchers studied the properties of learned subspace models such as convexity and dimension [2,24,25]. However, these approaches require many training images acquired under different lighting variations to learn the required low-dimensional subspaces. Finally, deep learning-based methods such as VGGface [10] and others [11–13] have become increasingly popular. These typically require a large amount of data for training. To overcome these limitations, many data augmentation strategies to increase number of training samples can be used [14]. More recently, Miao et al. proposed a knowledge-guided representation disentanglement network to learn face-related and illumination-related representations and then only utilized the face-related features for face recognition [26]. Huang et al. developed a deep face recognition framework, including a feature restoration network to generate illumination-enhanced images, a feature extraction network for computing face feature embeddings, and an embedding matching module for face classification [27]. Similarly, Fan et al. proposed a novel neural network first to enhance low-lighting face images and then employed a face recognition network for face classification [28]. However, when a limited amount of training data is available, even with data augmentation techniques [14], deep learning-based methods tend to have relatively poor generalization properties [29].

3. Proposed solution

We propose a novel approach for illumination invariant face recognition. The method splits the image into multiple image patches and

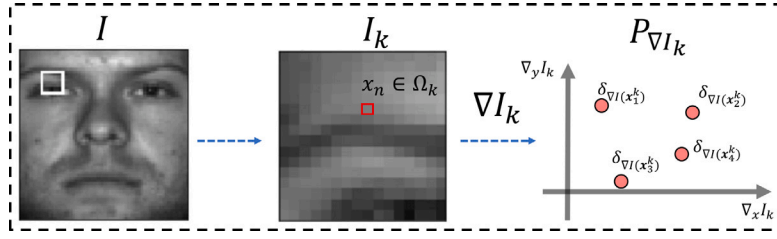


Fig. 2. From image space to local 2d discrete gradient distribution: given an image patch I_k , one can compute its 2d discrete gradient distribution $P_{\nabla I_k}$. Please note that here an illustration example is plotted to show $P_{\nabla I_k}$.

uses the local 2D discrete gradient distribution to represent the information in each patch. We demonstrate that illumination variations cause deformations of the local 2D discrete gradient distributions that can be modeled mathematically as well as learned from available data. Based on this observation, we propose a novel local image descriptor based on the Radon Cumulative Distribution Transform (R-CDT) [22] of the local gradient distribution. Compared with the continuous formulation of R-CDT in image space [22], the new proposed image descriptor is for local discrete gradient distributions within image patches. We model the set of all local image patches observed under certain illumination variations as a subspace in Discrete R-CDT transform domain. Given a test patch, the Discrete R-CDT transform of the test patch is computed, and the distance to the corresponding patch subspace for each subject in the database is computed. The average nearest subspace subject is then chosen as the classification result.

3.1. Effects of varying illumination on local 2D gradient distributions

3.1.1. Notation

An image $I : \Omega \rightarrow \mathbb{R}_+$ can be viewed as a mapping from the unit square $\Omega = [0, 1] \times [0, 1]$ to the set $\mathbb{R}_+$ of non-negative real numbers. Let $\Omega_k \subset \Omega$ refer to a set of pixel coordinates in the k th neighborhood in Ω and $\Omega_k^N = \{\mathbf{x}_1^k, \dots, \mathbf{x}_N^k\} \subset \Omega$ is the set of N pixel locations for patch k .

Given an image patch $I_k : \Omega_k \rightarrow \mathbb{R}_+$, the corresponding 2D discrete gradient distribution $P_{\nabla I_k}$ for the k th patch is defined as $P_{\nabla I_k} := \frac{1}{N} \sum_{\mathbf{x} \in \Omega_k^N} \delta_{\nabla I(\mathbf{x})}$. An example image I , image patch I_k , and corresponding distribution $P_{\nabla I_k}$ are shown in Fig. 2. Note our use of measure theoretic notation δ_z for a delta mass positioned at z . Section 7.2 in the supplementary material provides a summary of notations used in this paper as well as a brief introduction to Dirac measures.

3.1.2. Effects of illumination changes on 2D gradient distributions

An example of one image patch observed under different illumination conditions is shown in Fig. 1. The example shows that the gradient distribution $P_{\nabla I_k}$ undergoes deformations that may include translation, scaling, skewing, rotation, and nonrigid deformations according to the reflectance properties of the object being imaged, its three dimensional configuration, and specific illumination conditions. Here we propose a transport-based way of modeling such deformations.

Given the local gradient distribution $P_{\nabla I_k}$ of a subject observed under a certain illumination condition, the corresponding patch gradient distribution of the same subject after some illumination changes can be modeled by a bijective transformation h as: $P_{\nabla I_k}^h := h_{\#} P_{\nabla I_k} = \frac{1}{N} \sum_{\mathbf{x} \in \Omega_k^N} \delta_{h(\nabla I(\mathbf{x}))}$, $h \in \mathbb{H}$, where $\mathbb{H}$ is a set of bijective deformations from $\mathbb{R}^2$ to $\mathbb{R}^2$ that are particular to certain illumination assumptions. We denote the push forward of distribution $P_{\nabla I_k}$ as $h_{\#} P_{\nabla I_k}$ (see Section 7.4.2 in the supplementary material for definitions and properties of the push-forward operation). Moreover, we model the set of gradient distributions of a subject image patch I_k under various illumination variations with the following template-deformation based generative model: $\mathbb{P}_{\mathbb{H},k} = \{P_{\nabla I_k}^h \mid h \in \mathbb{H}\}$. A particular model for set $\mathbb{H}$ that describes illumination effects is proposed below. Here $\mathbb{P}_{\mathbb{H},k}$ refers to

the set of all possible observations of $P_{\nabla I_k}^h$. We will utilize geometric properties of $\mathbb{P}_{\mathbb{H},k}$ in our problem statement and solution described below.

3.1.3. Modeling $\mathbb{H}$: local illumination changes and patch generative model

The set $\mathbb{H}$ of deformations corresponding to the set of illumination variations within a subject class can be hard to specify in general. However, we can make some reasonable assumptions and propose an approximate model that can enable robust classification. Let the size (area) of neighborhood Ω_k be denoted as $|\Omega_k|$. We then postulate that, for small enough neighborhood size $|\Omega_k|$, we have that $h(\mathbf{z}) \sim \alpha \mathbf{z} + \mathbf{b}$ where $\mathbf{z}$ is a gradient coordinate (i.e. $\mathbf{z} = \nabla I_k(\mathbf{x})$, $\mathbf{x} \in \Omega_k$), and where $\alpha \in \mathbb{R}$ is an unknown scaling function, and $\mathbf{b} \in \mathbb{R}^2$ is an unknown translation vector. The illumination model above can be derived from the assumption that within a local patch, illumination variations within that patch can be expressed as

$$\alpha I_k(\mathbf{x}) + \beta + \mathbf{b}^T \mathbf{x}, \mathbf{x} \in \Omega_k. \quad (1)$$

under the assumption that $|\Omega_k|$ is small. In the equation above, $\alpha > 0$ is known as contrast, β illumination intensity, and $\mathbf{b}$ is a linear gradient (caused by illumination at an angle, or potentially shadows) superimposed on the image. Under the assumption of small neighborhood $|\Omega_k|$, the model $h(\mathbf{z}) \sim \alpha \mathbf{z} + \mathbf{b}$ can be understood as the gradient of (1), and thus can be understood in terms of illumination intensity, contrast, and linear illumination gradient. We note that similar smoothness assumptions are widely used for illumination-invariant face recognition [3–6,19].

Based on the model $h(\mathbf{z}) \sim \alpha \mathbf{z} + \mathbf{b}$ above, we propose a specific model for the set of bijections $\mathbb{H}$ that cause gradient deformations (push-forward) of a given gradient distribution: $\mathbb{H}_0 = \{h(\mathbf{z}) = \alpha \mathbf{z} + \mathbf{b} \mid \alpha > 0, \mathbf{b} \in \mathbb{R}^2\}$. Finally, using the notation established earlier, we express the set of gradient distributions observed under an unknown illumination function $h \in \mathbb{H}_0$ as $P_{\nabla I_k}^h := \frac{1}{N} \sum_{\mathbf{x} \in \Omega_k^N} \delta_{h(\nabla I(\mathbf{x}))} = \frac{1}{N} \sum_{\mathbf{x} \in \Omega_k^N} \delta_{\alpha \nabla I(\mathbf{x}) + \mathbf{b}}$, and the set of all possible observations as

$$\mathbb{P}_{\mathbb{H}_0,k} = \{P_{\nabla I_k}^h = h_{\#} P_{\nabla I_k} \mid h \in \mathbb{H}_0\}. \quad (2)$$

We denote the set defined in Eq. (2) as a model for the gradient distribution of patch k under illumination model $\mathbb{H}_0$. Specifically $\mathbb{H}_0$ is capable of isotropically scaling and translating a given distribution $P_{\nabla I_k}$. Fig. 6 and 7 in Section 7.1 in the supplementary material provide visualizations of these variations. Leveraging the knowledge present in Eq. (1) and Eq. (2), we propose the following patch-wise affine generative model and classification problem using gradient distributions for face images under varying illumination conditions.

Problem statement: Let the local illumination-based generative model for images pertaining to subject (class) $c = 1, \dots, C$ be defined as:

$$\mathbb{S}^c = \left\{ I^{c,j} \mid \begin{array}{l} I_k^{c,j}(\mathbf{x}) = \alpha_k^j I_k^c(\mathbf{x}) + \beta_k^j + \mathbf{b}_k^j \cdot \mathbf{x}, \\ \mathbf{x} \in \Omega_k, \alpha_k^j > 0, \beta_k^j \in \mathbb{R}, \mathbf{b}_k^j \in \mathbb{R}^2, \\ k = 1, \dots, K \end{array} \right\}, \quad (3)$$

where I^c refers to an (unknown) template image for subject (class) c . This model defines an infinite set, whose elements $I^{c,j}$ can be

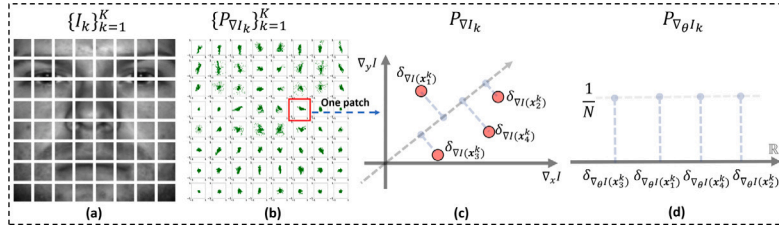


Fig. 3. An illustration example of computing a sliced one-dimensional discrete distribution for one image patch: (a) a face image consisted of K image patches $\{I_k\}_{k=1}^K$; (b) 2D discrete gradient distributions $\{P_{\nabla I_k}\}_{k=1}^K$ corresponding to image patches $\{I_k\}_{k=1}^K$; (c) a 2D discrete gradient distribution $P_{\nabla I_k}$ for a particular image patch; (d) sliced one-dimensional representation of $P_{\nabla I_k}$. Please note that here subplots (c) and (d) are only used for illustration purpose.

generated by applying the illumination model (1) on each patch k of I^c independently. In other words, the generative model is flexible to allow each patch k to contain its own contrast, brightness, and gradient vector $(\alpha'_k, \beta'_k, \mathbf{b}'_k)$ parameters, corresponding to different illumination effects (i.e. J_{th} illumination condition). These illumination parameters are unknown for any given photograph. Given L training images $\{I^{c,l}\}_{l=1}^L \subseteq \mathbb{S}^c$ from each class $c = 1, \dots, C$ where $\mathbb{S}^c \cap \mathbb{S}^{c'} = \emptyset$ for all $c \neq c'$, determine the class of an unknown image I' obtained from the same generative model.

3.2. Proposed solution

We propose a straightforward, non iterative, solution to the classification problem stated above. The solution is inspired on prior work on classification of distributions [21,22,30] and utilizes the fact that the gradient distribution $P_{\nabla I'_k}$ for patch k is an element of $\mathbb{P}_{\mathbb{H}_0,k}^c$, if $P_{\nabla I'_k}$ belongs to the class c . In other words $P_{\nabla I'_k} \in \mathbb{P}_{\mathbb{H}_0,k}^c$, with $\mathbb{P}_{\mathbb{H}_0,k}^c = \{h_{\theta} P_{\nabla I'_k} \mid h \in \mathbb{H}_0\}$, following the definition in Eq. (2), for some unknown c . We use a distance function $d(P_{\nabla I'_k}, \mathbb{P}_{\mathbb{H}_0,k}^c)$ that measures the Sliced Wasserstein Distance [31,32] between $P_{\nabla I'_k}$ and the nearest point in set $\mathbb{P}_{\mathbb{H}_0,k}^c$ to compute the solution of the classification problem stated above as:

$$c^* = \arg \min_c \sum_{k=1}^K d^2(P_{\nabla I'_k}, \mathbb{P}_{\mathbb{H}_0,k}^c). \quad (4)$$

It is easy to show that the minimization above obtains the correct solution to the problem statement above, provided that for at least one k we have that $\mathbb{P}_{\mathbb{H}_0,k}^c \cap \mathbb{P}_{\mathbb{H}_0,k}^{c'} = \emptyset$ whenever $c \neq c'$.

Below we show how we can estimate $d(P_{\nabla I'_k}, \mathbb{P}_{\mathbb{H}_0,k}^c)$ with the aid of a newly introduced operation which we will call the Discrete Radon Cumulative Distribution transform (Discrete R-CDT).

3.2.1. Sliced-Wasserstein representation of local 2D discrete distribution

Given an image I where the patch-wise representation is $\{I_k\}_{k=1}^K$, we compute its local gradient distribution $P_{\nabla I_k}$ as defined in Section 3.1.1 and use a modified version of the Radon Cumulative Distribution Transform (R-CDT) to represent $P_{\nabla I_k}$. The idea is to use a set of one-dimensional discrete distributions by “slicing” $P_{\nabla I_k}$ along different angles (see Fig. 3).

To begin, we note that one can compute the 1D distribution $P_{\nabla I_k}$ of projected gradients via

$$P_{\nabla I_k} = \frac{1}{N} \sum_{\mathbf{x} \in \Omega_k^N} \delta_{\nabla I(\mathbf{x}) \cdot \mathbf{w}_{\theta}}, \quad (5)$$

where $\mathbf{w}_{\theta} = (\cos \theta, \sin \theta)^T$ is a unit vector in the direction of θ . Inspired on earlier work on the CDT [30] and R-CDT [21], we then define the *Discrete CDT Transform* $\mathcal{F}$ for one-dimensional discrete probability distributions. Section 7.3 in the supplementary material provides the full definition. $\mathcal{F}$ takes the one-dimensional discrete distribution $P_{\nabla I_k}$ as input and outputs a vector which are concentration locations of the

discrete distribution in an increasing order:

$$\mathcal{F}(P_{\nabla I_k}) := P_{\theta} \begin{bmatrix} \nabla_{\theta} I(\mathbf{x}_1^k) \\ \nabla_{\theta} I(\mathbf{x}_2^k) \\ \vdots \\ \nabla_{\theta} I(\mathbf{x}_N^k) \end{bmatrix} = \begin{bmatrix} \nabla_{\theta} I(\mathbf{x}_1^k) \\ \nabla_{\theta} I(\mathbf{x}_2^k) \\ \vdots \\ \nabla_{\theta} I(\mathbf{x}_N^k) \end{bmatrix}, \quad (6)$$

where $\nabla_{\theta} I(\mathbf{x}_i^k) = \nabla I(\mathbf{x}_i^k) \cdot \mathbf{w}_{\theta}$ and P_{θ} is a permutation matrix such that $\nabla_{\theta} I(\mathbf{x}_i^k) \leq \nabla_{\theta} I(\mathbf{x}_{i+1}^k) \leq \dots \leq \nabla_{\theta} I(\mathbf{x}_N^k)$. In addition, one can show that $\mathcal{F}$ is an isometric embedding from the 1D discrete probability distribution space with the Wasserstein metric to the transform space with the Euclidean distance [31,33]. That is, given two 1D discrete distributions $P_{Z(1)}$ and $P_{Z(2)}$ in $\mathbb{P}(\mathbb{R})$, we have that the Wasserstein distance between them is given by $W_2(P_{Z(1)}, P_{Z(2)}) = \sqrt{\frac{1}{N} \|\mathcal{F}(P_{Z(1)}) - \mathcal{F}(P_{Z(2)})\|^2}$, where $\|\cdot\|$ denotes the Euclidean distance on $\mathbb{R}^N$.

Combining the “slicing” idea shown in (5) and the Discrete CDT Transform $\mathcal{F}$, we define the *Discrete R-CDT transform* $\mathcal{F}^*$ for 2D discrete distributions: $(\mathcal{F}^*(P_Z))(\theta) = \mathcal{F}(P_{Z_{\theta}})$ where $P_{Z_{\theta}} = \frac{1}{N} \sum_{i=1}^N \delta_{Z_i \cdot \mathbf{w}_{\theta}}$ is a one-dimensional distribution concentrated on the projected values of Z onto the directional vector $\mathbf{w}_{\theta}$. Section 7.5 in the supplementary material provides the complete description.

With the isometric embedding property in mind, we define the Discrete Sliced Wasserstein Distance [32] as $d(P_{Z(1)}, P_{Z(2)}) = (\int_0^{\pi} \|\mathcal{F}(P_{Z_{\theta}^{(1)}}) - \mathcal{F}(P_{Z_{\theta}^{(2)}})\|^2 d\theta)^{\frac{1}{2}}$, where $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^N$. In particular, when P_Z is a gradient distribution, say $P_Z = P_{\nabla I_k}$, then $\hat{P}_{\nabla I_k}(\theta) = (\mathcal{F}^*(P_{\nabla I_k}))(\theta) = \mathcal{F}(P_{\nabla I_k})$. In other words, applying the Discrete R-CDT (operator $\mathcal{F}^*$) on the local 2D discrete distribution (i.e. $P_{\nabla I_k}$) equates to applying Discrete CDT Transform (i.e., $\mathcal{F}$) on a collection of projected one-dimensional representations¹ of (e.g., $P_{\nabla I_k}$), indexed by $\theta \in [0, \pi)$. The corresponding Discrete Sliced Wasserstein distance between $P_{\nabla I'_k}$ and $P_{\nabla I_k}$ is $d(P_{\nabla I'_k}, P_{\nabla I_k}) = (\int_0^{\pi} \|\mathcal{F}(P_{\nabla I'_k}) - \mathcal{F}(P_{\nabla I_k})\|^2 d\theta)^{\frac{1}{2}}$.

The minimization problem (4) is hence equivalent to:

$$c^* = \arg \min_c \sum_{k=1}^K d(P_{\nabla I'_k}, \mathbb{P}_{\mathbb{H}_0,k}^c) = \arg \min_c \sum_{k=1}^K \min_{h_k \in \mathbb{H}_0} \|\hat{P}_{\nabla I'_k} - \hat{P}_{\nabla I_k}^{h_k}\|_{L^2([0,\pi), \mathbb{R}^N)} \quad (7)$$

In practice, we take θ from a finite set $\{\theta_1, \dots, \theta_m\}$ for some positive integer m , $\hat{P}_{\nabla I_k} = \mathcal{F}^*(P_{\nabla I_k})$ can be represented by a matrix of size $N \times m$ and reshaped as a long vector of length $m \cdot N$.

3.2.2. Nearest subspace learning

Next we leverage the generative model as in (3) together with the Discrete R-CDT transform $\mathcal{F}^*$ to form a nearest subspace classification method to facilitate the classification strategy in (4). It is not hard to see that $\mathcal{F}^*(\mathbb{P}_{\mathbb{H}_0,k}^c)$ is convex, meaning that $\lambda \mathcal{F}^*(P_{\nabla I_k}^{h_1}) + (1-\lambda) \mathcal{F}^*(P_{\nabla I_k}^{h_2})$ lies in $\mathcal{F}^*(\mathbb{P}_{\mathbb{H}_0,k}^c)$ for all $\lambda \in [0, 1]$ and $h_1, h_2 \in \mathbb{H}_0$. Section 7.6.1 in the

¹ This projection is similar to the sliced projections in the Radon transform, and this is why we have “Radon” (R) in the name of this new transform.

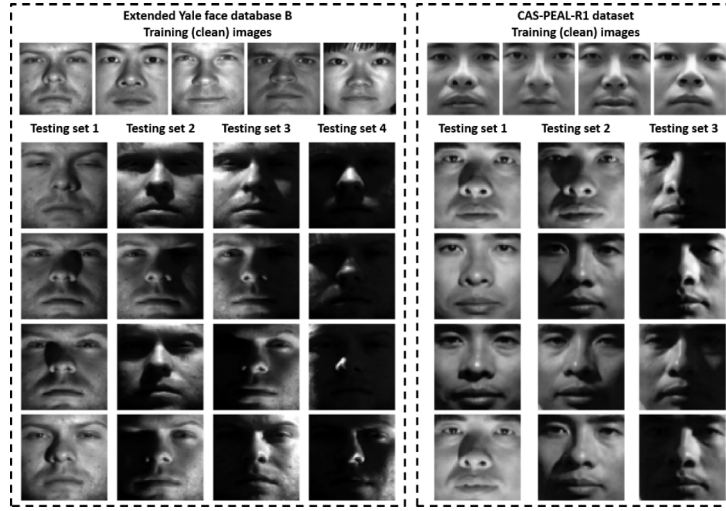


Fig. 4. Clean image examples from multiple subjects: (1) Extended Yale face database B (upper row); (2) CAS-PEAL-R1 dataset (bottom row).

supplementary material has the proof. Indeed, the deformations $\mathbb{H}_0$ in 2D discrete distribution space also cause the corresponding translation and scaling effects in Discrete R-CDT transform space:

$$\mathcal{F}^*(P_{(\alpha \nabla I_k + \mathbf{b})})(\theta) = \alpha \mathcal{F}^*(P_{\nabla I_k})(\theta) + \mathbf{b} \cdot \mathbf{w}_\theta, \quad (8)$$

using the composition property of the Discrete CDT transform (please see Section 7.4.2 in the supplementary material section). We then expand the convex set $\mathcal{F}^*(\mathbb{P}_{\mathbb{H}_0, k})$ to form a subspace $\mathbb{V}_k^c = \text{span}(\mathcal{F}^*(\mathbb{P}_{\mathbb{H}_0, k}^c))$ and further assume that when $c \neq c'$, there exists a patch k such that $\mathbb{P}_{\mathbb{H}_0, k}^c \cap \mathbb{V}_{k'}^{c'} = \emptyset$, which is consistent with the assumption that the image class of subject c will not overlap with images of a different subject c' under all possible illumination variations. With the above considerations in mind, the constrained minimization problem in (4) can be modified to a subspace projection problem in transform domain as: $c^* = \arg \min_c \min_{\hat{P} \in \mathbb{V}_k^c} \sum_{k=1}^K \|\hat{P}_{\nabla I_k'} - \hat{P}\|_{L^2((0, \pi), \mathbb{R}^N)} = \arg \min_c \sum_{k=1}^K d(\hat{P}_{\nabla I_k'}, \mathbb{V}_k^c) = \arg \min_c \sum_{k=1}^K d_k^c$, where $d_k^c := d(\hat{P}_{\nabla I_k'}, \mathbb{V}_k^c)$ is the $d(\cdot, \cdot)$ distance of $\hat{P}_{\nabla I_k'}$ to the subspace $\mathbb{V}_k^c$, which can be computed using a least squares projection as shown below. In summary, the class is determined by the smallest distance $d^c = \sum_{k=1}^K d_k^c$. In particular, if a test image I' belongs to subject c , $\sum_{k=1}^K d_k^c < \sum_{k=1}^K d_k^{c'}$ for any $c' \neq c$ (see Section 7.7 in the supplementary material for more details).

3.3. Implementation

We detail the implementation of training and testing procedures in the subsequent paragraphs.

Training. Given a total of $\{I_l^c\}_{l=1}^L$ L training images for c_{th} subject. Each image I_l is partitioned into K image patches $\{I_{l,k}^c\}_{k=1}^K$. We approximate the subspace $\mathbb{V}_k^c$ using L training images $\{I_l^c\}_{l=1}^L$ by $\mathbb{V}_k^c = \text{span}(\{\hat{P}_{\nabla I_{l,k}^c}, \dots, \hat{P}_{\nabla I_{L,k}^c}\} \cup \mathbb{U}_T)$, where $\mathbb{U}_T = \{\mu_1(n, \theta), \mu_2(n, \theta)\}$ with $\mu_1(n, \theta) = \cos \theta$, $\mu_2(n, \theta) = \sin \theta$ can be used to automatically model translation and scaling within a subject gradient distribution class by observing Eq. (8) [22,29]. It is worth noting that if the deformation strictly follows the set $\mathbb{H}_0$, only taking span of one transformed training example with $\mathbb{U}_T$ is necessary. However, in reality it is often the case that more complicated illumination effects than defined in $\mathbb{H}_0$ are present. We can enhance $\mathbb{H}_0$ by using any available training images to take span of multiple transformed training examples with $\mathbb{U}_T$. This technique allows the proposed method to learn from more complicated lighting variations. A more detailed discussion is provided in Section 5.2.

Recall that in our algorithm, each $\hat{P}_{\nabla I_{l,k}^c}$ in the spanning set can be discretized into a vector in $\mathbb{R}^{mN}$ where m is number of angles used in

practice (see Remark 3.1). For simplicity, we abuse the notation and use the same symbol for the discretized version of $\hat{P}_{\nabla I_{l,k}^c}$ in algorithms, which will be clear from the context. Similarly one can discretize functions in $\mathbb{U}_T$ by evaluation on the grid $\{1, 2, \dots, N\} \times \{\theta_1, \dots, \theta_m\}$ and reshape them as a vectors of length $m * N$, which we abuse notation and denote as $\mathbb{U}_T$. Now we summarize the training algorithm in the following steps: for each class c and each patch k ,

1. Compute the transforms $\hat{P}_{\nabla I_{1,k}^c}, \dots, \hat{P}_{\nabla I_{L,k}^c}$ corresponding to the training images
2. Use Principal Component Analysis (PCA), keeping enough components to retain 99% of the training data variance, to orthogonalize $\{\hat{P}_{\nabla I_{1,k}^c}, \dots, \hat{P}_{\nabla I_{L,k}^c}\} \cup \mathbb{U}_T$ to obtain a set of orthonormal basis vectors $\{v_{1,k}^c, v_{2,k}^c, \dots\}$ and form a matrix B_k^c with $\{v_{1,k}^c, v_{2,k}^c, \dots\}$ as its columns: $B_k^c = [v_{1,k}^c, v_{2,k}^c, \dots]$. When enough training data is available, we split the training data into training and validation sets, and choose the smallest number of components that allow for highest classification accuracy on the validation set.

Testing. Given a testing image I' , the first step is to segment I' into K image patches $\{I_{l,k}'\}_{k=1}^K$ and calculate the Discrete R-CDT transform representation $\{\hat{P}_{\nabla I_{l,k}'}\}_{k=1}^K$. Then, for each image patch k , we compute the distance $d(\hat{P}_{\nabla I_{l,k}'}, \mathbb{V}_k^c) = \|\hat{P}_{\nabla I_{l,k}'} - B_k^c (B_k^c)^T \hat{P}_{\nabla I_{l,k}'}\|$, where $(B_k^c)^T$ is the transpose of matrix B_k^c and $\|\cdot\|$ denote the Euclidean norm. Finally, we compute d^c by summing the distance contribution d_k^c and search for the nearest subspace as the classification result via $\arg \min_c d^c = \sum_{k=1}^K d_k^c$.

4. Experiments

We evaluate the proposed method on three of the largest face illumination datasets: (1) Extended Yale Face Database B [2]; (2) AR Face Dataset [34]; (3) CAS-PEAL dataset [35]. The task is to perform face recognition.

The extended yale face database B. contains 38 subjects, each of which has 64 images under different illumination conditions.² Based on the degree of lighting directions, for each subject, we split the dataset into five subsets as commonly done [4–6]: in *training subset*, each subject has 11 images, in *test subset 1*, each subject has 10 images, in *test subset 2*, each subject has 18 images, in *test subset 3*, each subject has 12 images, and in *test subset 4*, each subject has 13 images. Each image

² Please note that several subjects do not have 64 images due to corrupted images during the acquisition phase as indicated in [2].



Fig. 5. Image examples from AR face dataset: (1) Top row shows clean images from multiple subjects; (2) second to fourth row show images with illumination variations (e.g., right light on, left light on, and both sides light on).

has a distinct lighting condition. In Fig. 4, the left panel shows images in the *training subset* for multiple subjects and several testing samples from *test subset 1* to *test subset 4*.

CAS-PEAL-R1 dataset. contains 233 different subjects. Each subject only has one “clean” (with standard illumination) image and at least 9 images with varying lighting effects. Based on the azimuth angle of light direction, we divide the face images with various levels of illumination into three different subsets: *Test subset 1* where azimuth angle equals to 0 degree, *Test subset 2* where azimuth angle is either -45 or 45 degree, and *Test subset 3* where azimuth angle is either -90 or 90 degree. Fig. 4 right panel illustrates training and testing images from the CAS-PEAL-R1 dataset.

Arface dataset. contains 100 subjects with mild illumination variations as shown in Fig. 5. Each subject has two sets of images. Each set contains one “clean” image and 3 images with different levels of illumination. In total, each subject has 2 clean images and 6 images with different lighting variations. We combine images with left and right illumination on as the *test subset 1*, while images with side lighting on are included for *test subset 2*. Similarly, each model is trained using 2 clean images, and is tested on *test subset 1* and *test subset 2* respectively.

We perform comparisons between the proposed method and other illumination-invariant face recognition algorithms [3–6,16,23] as well as several deep learning based alternatives [11–13] with illumination data augmentation strategy. Specifically, we consider three state-of-the-art deep learning models: VGGFace, ResNet-50 and DenseNet-121. We use 90% and 10% of the original training data for training and validation, respectively. Validation is performed every ten iterations, the final test accuracy is based on the model checkpoint that has the best validation accuracy. When there is only one training sample available, each sample is augmented 5 times using the illumination model stated in Eq. (1). For all the experiments we use an Adam optimizer with a learning rate of 0.001.

4.1. Results

Two experiments, denoted as *Test 1* and *Test 2*, were conducted to evaluate the proposed approach using different training strategies for this dataset. *Test 1* trains the proposed model by utilizing all images in the *training set*, and performs evaluations on *Test subsets 1, 2, 3, and 4*, respectively. We further remark that in *Test 1*, since there are multiple training images available, we set aside a validation set to choose the number of components in the orthogonalization step (please see Section 3.2.2) using PCA. Specifically, We use a 66% and 33% of original training data for training and validation. We use the validation set to define the smallest number of components that maximize

classification accuracy. *Test 2* trains the proposed model using one single sample, which is randomly selected from the *training set*, then it follows the same testing procedure as discussed above in *Test 1*. Table 1 presents experimental evaluation results of *Test 1* and *Test 2*. Clearly, the proposed method achieves the top and most robust performance across different test subsets in both of *Test 1* and *Test 2*. Specifically, in *Test subset 1 and 2* of *Test 1*, most of the approaches achieve satisfactory performance, meaning that they are able to address mild illumination effects for face recognition tasks. With increasing levels of illumination conditions in *Test subset 2,3, and 4*, such as when uneven and large areas of shadow are present, other methods experienced significant decreases in performance. A similar trend can be observed in *Test 2* as well. Of the comparison methods, the WebberFace, GradientFace, and MSLDE approaches consistently achieve better performance than their deep learning counterparts, indicating that they are able to eliminate some illumination effects after the “normalization” stage, especially in the test subsets (e.g., *Test subset 1 and 2*) with low-level of illumination variations. In addition, incorporating the log transform into HoG + SVM pipeline also improves performance.

Regarding the deep learning-based methods, some achieve excellent performance in *Test subset 1 and 2* of *Test 1* and *Test 2*, where the minimum illumination effects exist. Nevertheless, for *Test subset 3 and 4*, their classification accuracy decrease significantly. This may be due to the fact that only a limited number of training samples are available. As commonly done in the machine learning literature, we employed a data augmentation strategy to address the shortage of training data. Specifically, we adopt the model stated in Eq. (1) to randomly augment training samples with different levels of illumination (e.g., to simulate various contrast and brightness changes).³ Each parameter in Eq. (1) is randomly configured as $\alpha \in [0.1, 3]$, $\beta \in [1, 30]$, $\mathbf{b} \in [0.1, 3]$. We observe that for deep learning approaches illumination data augmentation strategy improves classification performance to a degree in cases where illumination changes are small (e.g., in *Test subset 1 and 2*). However, when illumination effects become severe, e.g., in *Test subset 3 and 4*, data augmentation become less effective. Though using data augmentation is a valid option to increase performance, it is still extremely difficulty to prescribe how to augment the data, in addition to other issues such as computational complexity and out of distribution performance issues [29]. Finally, we note that the performance of deep learning-based approaches can vary significantly due to the network architecture.

³ Please note in here equation (1) is applied to the entire image domain rather than a single patch.

Table 1

Performance comparison for the extended Yale face database.

	Extended Yale face database (Number of classes = 38)							
	Test 1				Test 2			
	Test subset 1	Test subset 2	Test subset 3	Test subset 4	Test subset 1	Test subset 2	Test subset 3	Test subset 4
WebberFace + SVM	100%	95.1%	94.2%	90.6%	92.1%	78.8%	70.1%	74.4%
MSLDE + SVM	100%	93.3%	82.0%	79.1%	87.6%	64.9%	60.8%	64.0%
GradientFace + SVM	100%	91.5%	88.4%	85.1%	80.5%	58.6%	65.2%	60.0%
Log + DCTface + SVM	97.8%	80.5%	70.7%	43.6%	72.8%	41.3%	39.8%	24.6%
HoG + SVM	100%	82.4%	49.1%	65.9%	68.6%	47.2%	30.7%	42.0%
Log + HoG + SVM	100%	91.1%	70.5%	80.6%	73.1%	51.9%	44.2%	51.0%
VGGface	67.8%	14.6%	3.7%	2.4%	15.0%	5.7%	2.4%	2.6%
VGG face with data aug.	98.6%	77.1%	55.5%	25.3%	66.0%	33.5%	10.6%	7.7%
ResNet-50	96.5%	50.3%	9.5%	4.0%	63.9%	14.9%	3.5%	2.2%
ResNet-50 with data aug.	97.8%	73.3%	42.0%	26.9%	91.0%	49.3%	23.8%	14.4%
DenseNet-121	85.5%	36.4%	5.9%	3.8%	57.1%	15.5%	4.6%	2.0%
DenseNet-121 with data aug.	98.1%	58.7%	23.4%	14.8%	78.4%	24.8%	9.2%	2.0%
Discrete R-CDT + NS	100%	98.8%	96.2%	94.4%	98.4%	95.5%	92.4%	91.8%

Table 2

Performance comparison for the CAS-PEAL and ARface datasets.

	CAS-PEAL (Number of classes = 233)			ARface (Number of classes = 100)	
	Test subset 1	Test subset 2	Test subset 3	Test subset 1	Test subset 2
WebberFace + SVM	16.2%	19.5%	27.5%	93.7%	87.0%
MSLDE + SVM	18.0%	17.9%	40.0%	97.5%	94.5%
GradientFace + SVM	21.1%	20.0%	38.4%	97.5%	93.0%
Log + DCTface + SVM	11.4%	8.5%	4.0%	87.5%	73.0%
HoG + SVM	19.3%	13.2%	19.0%	99.7%	83.5%
Log + HoG + SVM	22.8%	17.2%	24.3%	100.0%	95.5%
VGGface	2.3%	1.7%	0.7%	19.7%	11.5%
VGG face with data aug.	5.5%	3.5%	2.6%	80.2%	58.4%
ResNet-50	3.8%	1.8%	1.0%	59.5%	7.5%
ResNet-50 with data aug.	6.9%	4.8%	5.1%	92.7%	80.0%
DenseNet-121	1.4%	1.1%	0.8%	54.2%	9.0%
DenseNet-121 with data aug.	3.3%	2.0%	1.0%	89.9%	82.4%
D-RCDT + NS	42.2%	50.2%	49.7%	99.5%	94.5%

In evaluation of CAS-PEAL-R1 dataset, the model only utilizes the single clean image of each subject for training. Evaluations are performed on three testing subsets: *Test subset 1, 2, and 3*. Table 2 left panel summarizes the experiment results. Overall, the proposed method outperforms other alternatives by a large margin, which is a consistent trend across the three testing subsets. With respect to the comparison methods, GradientFace and MSLDE achieve the best performance. Deep learning-based approaches are not able to perform well perhaps due to the limited number of training samples, even illumination data augmentation strategies are used.

For ARface dataset, Table 2 right panel provides the classification accuracy for *test subset 1 and 2*. We notice that the method Log + HoG [23] method achieves the best performance while the method proposed here obtains comparable performance. Other methods such as MSLDE and GradientFace also have high accuracies. It is worth noting that illumination data augmentation are most effective for the AR face dataset, which allows deep learning approaches to gain significant performance improvement, implying that employing data augmentation is an effective route to address mild illumination variations.

5. Discussion

5.1. Hyperparameter study

The experimental results presented above show that the face recognition approach we proposed is more robust to variations in illumination conditions than a variety of existing methods. The method is based on splitting face images into finite support neighborhoods Ω_k , and using a Discrete R-CDT representation for the gradient distribution within each neighborhood. As such, certain parametric choices (number of projections in R-CDT, neighborhood size, and neighborhood overlap) have to be made. To study the best choices for these parameters, we

followed the same training and testing strategy used in *Test 2* for the Yale Face dataset and focus on the *Test subset 4*, which has the most challenging illumination conditions. Results are present in the Table 4 in Section 7.8 (supplementary material). When varying different patch sizes, the overlap size is set as 0. Overall, results show that utilizing smaller patch improves the performance. We postulate this may be because the illumination model described in Eq. (1) is more accurate for small neighborhoods. Secondly, when the neighborhood size is fixed, increasing number of projections and overlap size are able to provide performance improvement to some extent.

5.2. Future work

The proposed approach above builds on prior work related to patch-wise analysis of gradient distributions. One of the novel contributions described in our method is the modeling of certain illumination transformations as transport operations on the resulting gradient distributions. Based on this point of view, there are multiple interesting perspectives worth further exploring as future work. Firstly, as discussed in Section 3.1, the log pixel transformation is a widely used technique to circumvent illumination effects as shown in prior work [3, 6, 23]. Based on this we also applied the log transform as a preprocessing step for our method. Table 5 in Section 7.8 (supplementary material) demonstrates classification results as well as the performance improvement (as highlighted using an upward arrow) for three face datasets used in the experiment. In general, employing the log transform increases classification accuracy of our method for most tests. In addition, the largest performance increase takes place in where the dataset has large regions of illumination variations such as *test subset 4* in the Yale Face dataset. We note that the application of the log transform on image space changes equation (1). Therefore, a careful mathematical analysis of incorporating the log transform in our

gradient distribution representation will be the subject of future work.

Secondly, the affine illumination generative model in Eq. (2) includes spatial transformations such as scaling and translation of the 2D discrete gradient distribution. Other 2D discrete gradient distribution deformations such as rotation, anisotropic scaling, and shearing, are worth exploring as future work in connection with more elaborate illumination models. Specifically, Shifat-E-Rabbi *et al.* [29] developed a mathematical formulation for rotation, anisotropic scaling, and shear deformations in the sliced-Wasserstein space. Future work will consider adopting the same idea to model rotation, anisotropic scaling, and shearing effects of 2D discrete gradient distribution in discrete R-CDT domain, if more complex illumination models require it.

We also note that when recognizing faces certain image patches contain more discriminative information than others. For example patches near the eyes, nose, and mouth, tend to contain more information humans use to identify one another. In this pilot study, when aggregating the distance d_k^c from each image patch, we implicitly assume that each image patch has a equal weight coefficient. However, we can rewrite the equation $\arg\min_c \sum_{k=1}^K d_k^c$ as $\arg\min_c \sum_{k=1}^K p_k d_k^c$, where p_k is the weight coefficient of k_{th} patch. This means that certain p_k s will have higher weight than others. Therefore, future study will investigate a learning-based approach to derive such weight coefficients p_k s for potential performance improvements.

Finally, we argue that taking the span of multiple transformed training examples (e.g., $\mathbb{V}_k^c = \text{span}(\{\hat{P}_{V_{I,k}^c}, \dots, \hat{P}_{V_{L,k}^c}\})$) allows the proposed model to capture more complicated lighting variations than the ones captured by the set $\mathbb{H}_0$. Table 1 demonstrates this by showing that an increasing number of training images, the classification accuracy improves as more data is added for each test subset. Furthermore, we conducted a similar experiment on the CAS-PEAL-R1 face dataset. Note that, originally, the CAS-PEAL-R1 face dataset only has one “clean” image for each subject. In order to expand the training set, we added images from *test subset 1* and *test subset 2* to the training set and evaluated the algorithm on *test subset 3*. Results shows that the classification accuracy increases from 49.7% to 70.17% when training images include *test subset 1* images, and to 82.51% when images from both of *test subset 1* and *test subset 2* are included for training, respectively. These results indicate the proposed method is learning gradient distribution transformations that go beyond $\mathbb{H}_0$ (scaling and translation). As such, we believe that enhanced methods for combining data (in transform domain) and illumination models $\mathbb{H}$ could potentially yield even higher accuracy results.

6. Conclusion

In conclusion, we approached the illumination invariant face recognition problem from a novel transport perspective by modeling spatial transformations in local image patches, such as the scaling and translation of the local 2D discrete gradient distribution. We mathematically demonstrated the local image gradient distributions form a convex set in the discrete R-CDT domain under certain illumination conditions. Leveraging this convexity allows us to use a nearest subspace classifier for illumination invariant face classification. The mathematical formulation can contribute to developing a more interpretable illumination-invariant face recognition algorithm. Furthermore, experimental results showed that the proposed method outperformed other alternatives in several face datasets with challenging lighting variations, especially when the number of training samples is limited. Modeling other more complex deformations or learning the deformations would be the future work for further performance improvement. Nevertheless, the proposed method, publicly available at https://github.com/rohdelab/drcdt_face, offers a novel and promising solution for illumination invariant face recognition.

CRedit authorship contribution statement

Yan Zhuang: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Shiying Li:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Mohammad Shifat-E-Rabbi:** Investigation, Methodology, Software, Writing – original draft, Writing – review & editing. **Xuwang Yin:** Investigation, Methodology, Writing – original draft, Writing – review & editing. **Abu Hasnat Mohammad Rubaiyat:** Investigation, Methodology, Writing – original draft, Writing – review & editing. **Gustavo K. Rohde:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patcog.2025.111381>.

Data availability

The dataset used in this study is publicly available. Python code implementing the proposed method is available at https://github.com/rohdelab/drcdt_face.

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Yan Zhuang received the MS degree from the State University of New York (SUNY) at Buffalo in 2014 and the Ph.D degree from the University of Virginia in 2022. He was a postdoctoral research fellow at Department of Radiology and Imaging Science, NIH Clinical Center. Currently he is a faculty instructor at Windreich Department of Artificial Intelligence and Human Health and Department of Diagnostic, Molecular and Interventional Radiology at Icahn School of Medicine at Mount Sinai. His research focuses on medical imaging analysis and machine learning.

Shiying Li received her Ph.D degree in mathematics in 2019 from Vanderbilt University, Nashville, Tennessee. She is currently an Assistant Professor in the Department of Mathematics at the University of Nebraska – Lincoln. Prior to her current position, she was a Postdoctoral Research Associate in the Department of Biomedical Engineering at the University of Virginia and later in the Department of Mathematics at the University of North Carolina - Chapel Hill. Her research focuses on applied optimal transport and mathematical data science.

Mohammad Shifat-E-Rabbi received the B.Sc. degree in electrical and electronic engineering from the Bangladesh University of Engineering and Technology, Dhaka, Bangladesh, in 2015. He is currently pursuing the Ph.D. degree in the Imaging and Data Science Laboratory of the Department of Biomedical Engineering, University of Virginia. His current research interests include applied mathematics, machine learning, data science, computational biology.

Xuwang Yin received his Ph.D degree in computer engineering from the University of Virginia. He is currently a research engineer at the Center for AI Safety (CAIS). His research interests include adversarial machine learning and generative modeling.

Abu Hasnat Mohammad Rubaiyat received his B.S. degree in electrical engineering from the Bangladesh University of Engineering and Technology in 2013, M.S. degree in electrical engineering from the West Virginia University in 2017, and his Ph.D. degree in Computer Engineering from the University of Virginia in 2023. His research is focused on applications of the transport-based transforms in signal and image processing applications.

Gustavo K. Rohde received the B.S. degree in physics and mathematics in 1999 and the M.S. degree in electrical engineering in 2001 from Vanderbilt University, Nashville, Tennessee. He received the doctorate in applied mathematics and scientific computation in 2005 from the University of Maryland, College Park. He is currently a Professor of biomedical engineering and electrical engineering at the University of Virginia, Charlottesville, where he leads the Imaging and Data Science Laboratory.